NEUROSTRIP USER GUIDE

NEUROSTRIP™

CONTROL BIONICS

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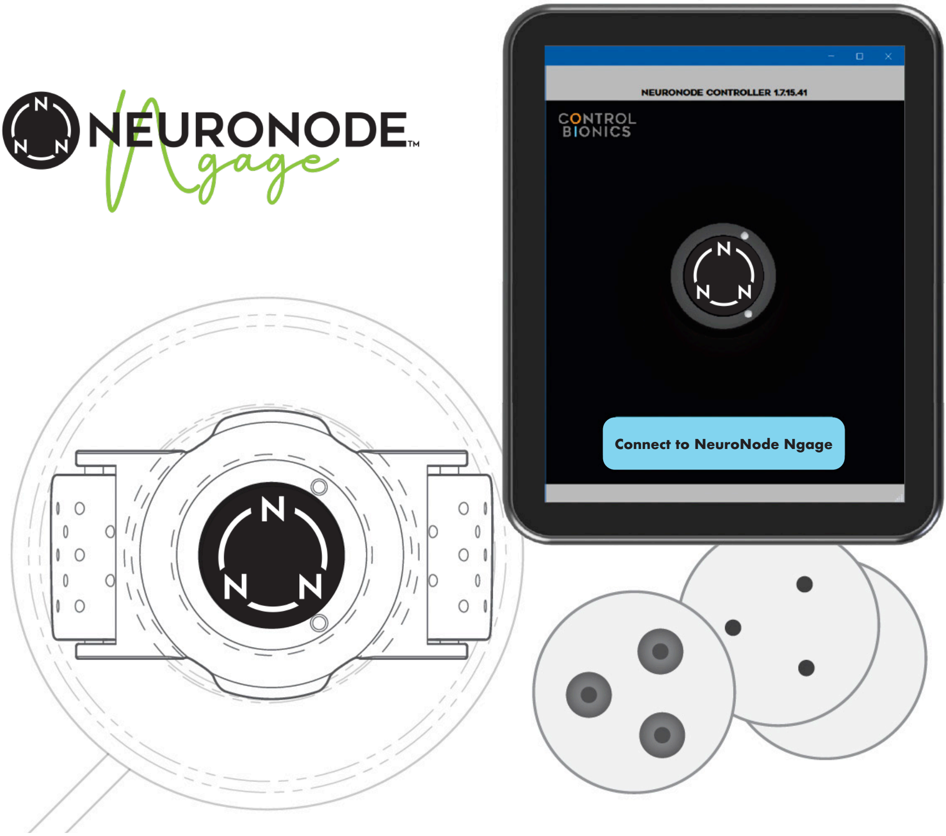
CONTENTS

1.0 WELCOME TO NEUROSTRIP

NeuroStrip offers insight into muscle activation and intent through bio-electric signals. Using high fidelity sensors, the patent-pending NeuroStrip measures and records surface electromyography (sEMG) data from the muscle and accelerometer data from device movement. Allowing it to be utilized across a multitude of industries including rehabilitation, sport science, wearables and medical technology.

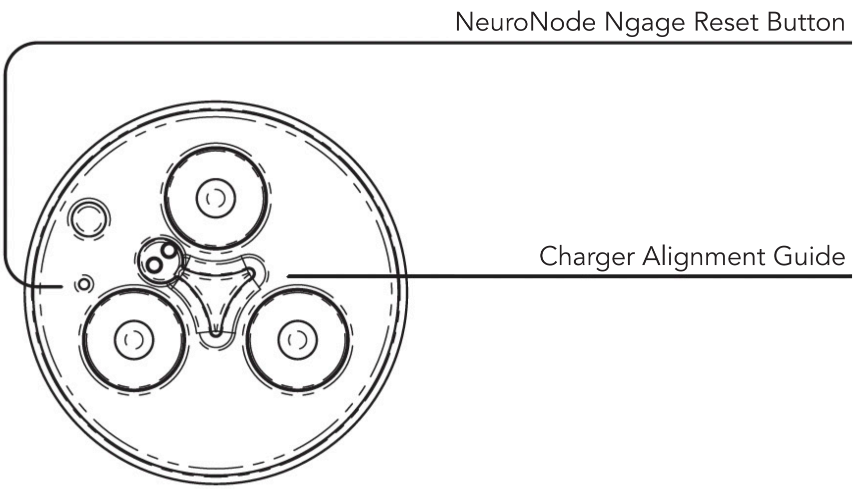
The NeuroStrip Application allows for multiple devices to be connected and analyzed in a single session allowing for more insight then ever before. With a variety of customizable options, the NeuroStrip Application allows for highly customized analysis. With the ability to record and review sessions within the application, track progress and further enhance client recovery and performance.

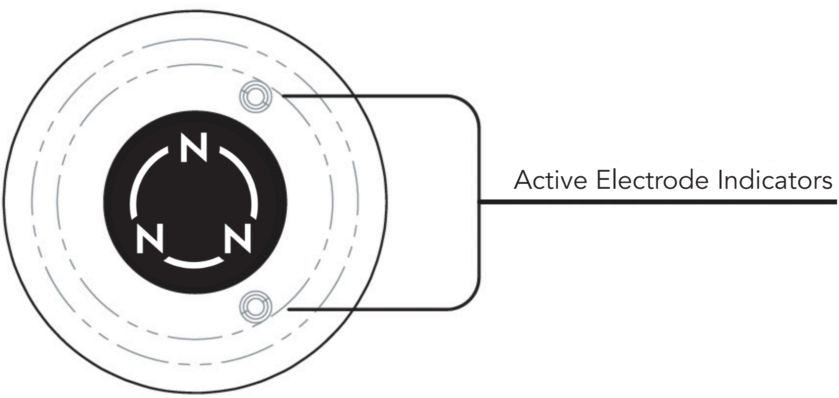


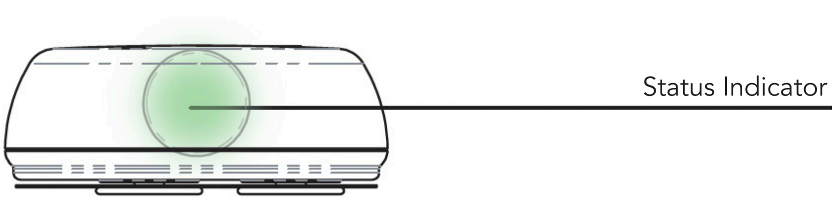
2.0 WELCOME TO NEURONODE NGAGE

NeuroNode Ngage offers insight into muscle activation and intent through bio-electrical signal data. As a wearable, wireless sEMG device, Ngage uses adhesive electrodes placed on the skin to capture accurate muscle fiber activity and support meaningful muscle activation tracking and analysis. As an alternative to NeuroStrip, Ngage gives users another flexible setup option for capturing muscle activation data in rehabilitation, sport science, strength and conditioning, wearables, and medical technology applications.

Muscle activation data is collected, displayed, and summarized through clear statistics, graphs, and performance insights. This helps clinicians, trainers, and strength and conditioning coaches review activation patterns, guide sessions, track progress, and identify areas for improvement.

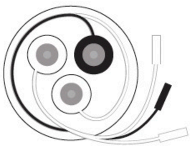

NeuroNode Ngage Reset Button
Charger Alignment Guide


Active Electrode Indicators

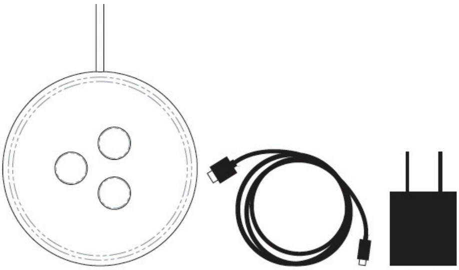

Status Indicator

2.2 OVERVIEW OF NEURONODE NGAGE ACCESSORIES

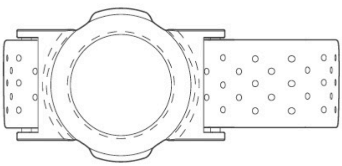
NeuroNode
Ngage



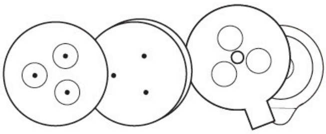
Lead-wire
Adaptor Base



Charger Kit

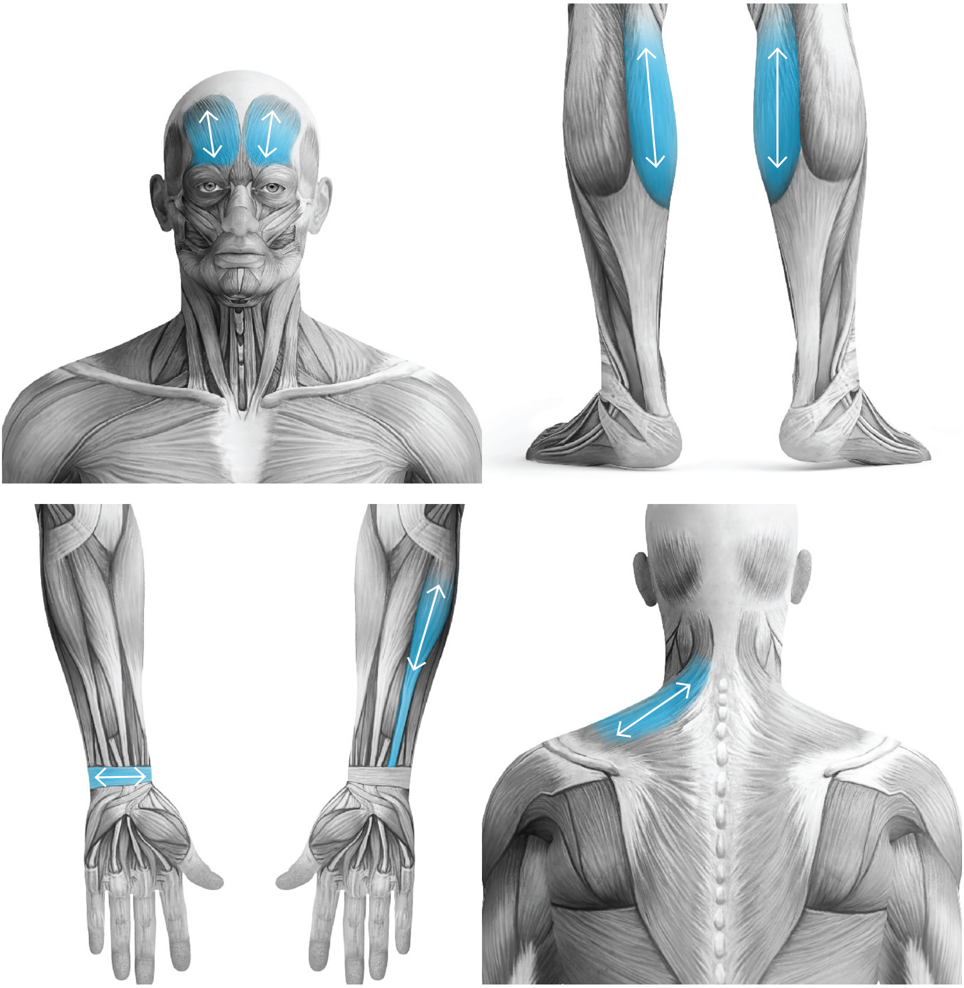


NeuroNode
Ngage Band



Electrode Kit

ITEM	FUNCTION
NEURONODE NGAGE	The NeuroNode Ngage is a wearable, wireless, EMG + 3D Spatial device.
LEAD-WIRE ADAPTER BASE	The Lead-wires connect to the Lead-wire Adapter Base of the NeuroNode Ngage. The three Snap-leads on the other end of the Lead-wires attach to the electrodes on the user's skin.
CHARGER KIT	The NeuroNode Ngage Charging Base and included cords allow for hassle-free charging for up to 24 hours of battery life.
NEURONODE NGAGE BAND	The NeuroNode Ngage Band is an adjustable, adhesive-free solution for placement of the NeuroNode Ngage on the user's skin.
ELECTRODE KIT	A starter electrode kit is included with the NeuroNode Ngage. This kit includes both adhesive and non-adhesive solutions as required.



3.0 EMG PLACEMENT

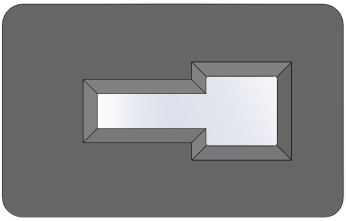
The chosen EMG Target Muscle should respond, at least minimally, to a command to contract the muscle. The muscle should return to a resting state in a timely manner.

The muscle does not need to function at optimal levels. The NeuroStrip is designed to detect muscle movement reliably and accurately to microvolt signal levels at the Target Muscle site.

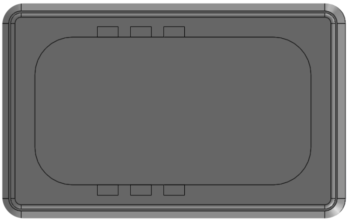
3.1 PATCH ALIGNMENT

The NeuroStrip kit includes the Patch Alignment Tool. This tool is used to properly align the NeuroStrip to the electrode targets on the patch.

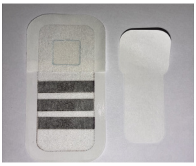
PATCH ALIGNMENT TOOL



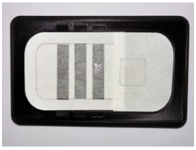
TOP



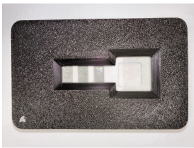
BASE



STEP 1:
Remove Key hole liner from top of NeuroStrip Patch.

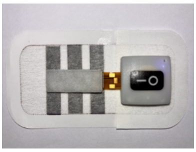


STEP 2:
Place patch in base of patch alignment tool.

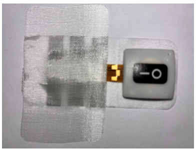


STEP 3:
With patch inserted, place top of the tool onto the base making sure it is a flush, snug fit.

STEP 4:
With tool together, insert NeuroStrip into outline and pressed down to adhere to the patch.



STEP 5:
Remove the joined NeuroStrip and patch from the tool.



STEP 6:
Remove the bottom liner, adhere to the body and apply the overdressing to the tail section only.

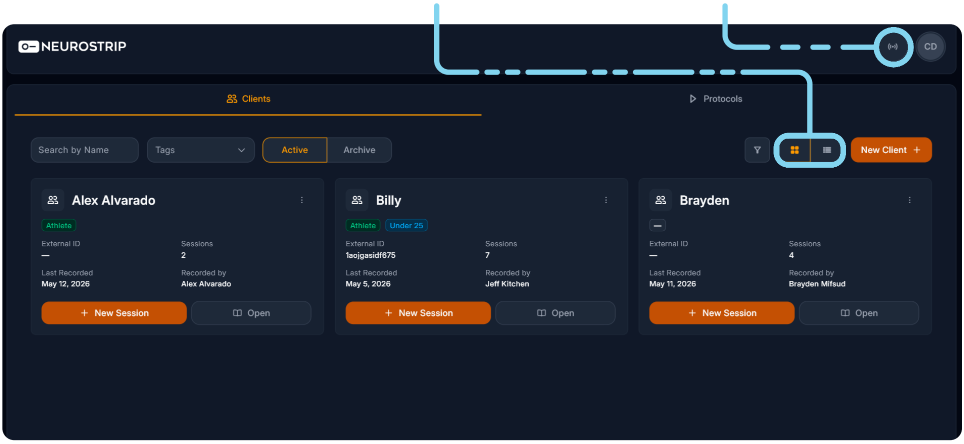
4.0 INTRO SCREEN

Once logged in, the Intro Screen serves as the main dashboard for accessing and managing clients and protocols, with tools to search, filter, create new records, start sessions, and open existing client or protocol details.

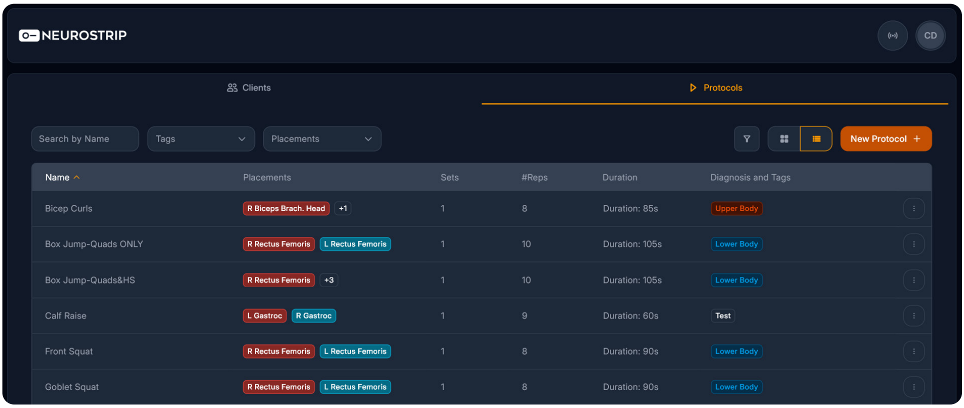
CLIENTS TAB:
Users can search and filter patient records, view active or archived patients, create a new patient, open an existing patient profile, and start a new session directly from a patient card.

LIST / GRID VIEW
Toggle between list/grid view.

CONNECTIONS
View connected devices.



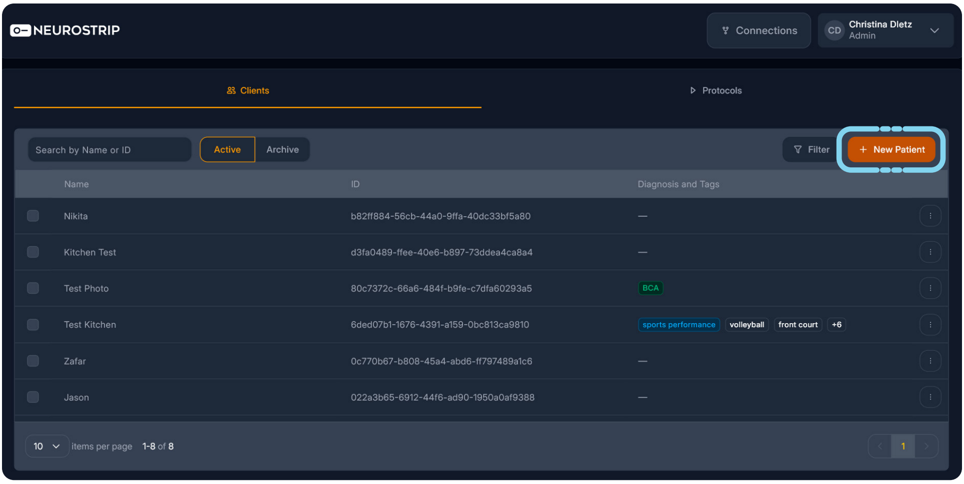
PROTOCOLS TAB:
On the Protocols tab, users can search and filter protocols, create a new protocol, review protocol details, and use the actions menu to manage existing protocols.

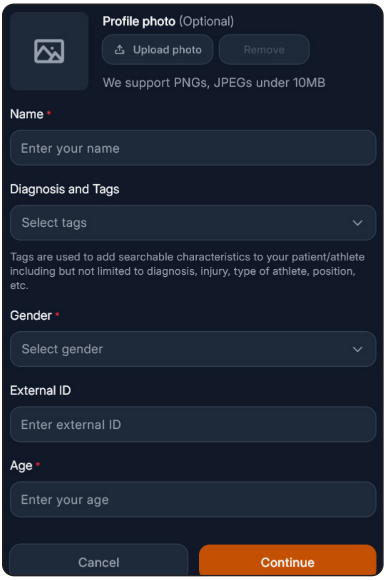


4.1 NEW PATIENT SETUP

The NeuroStrip Application lets you manage client records directly within the app. Add a new client or select an existing profile to begin.

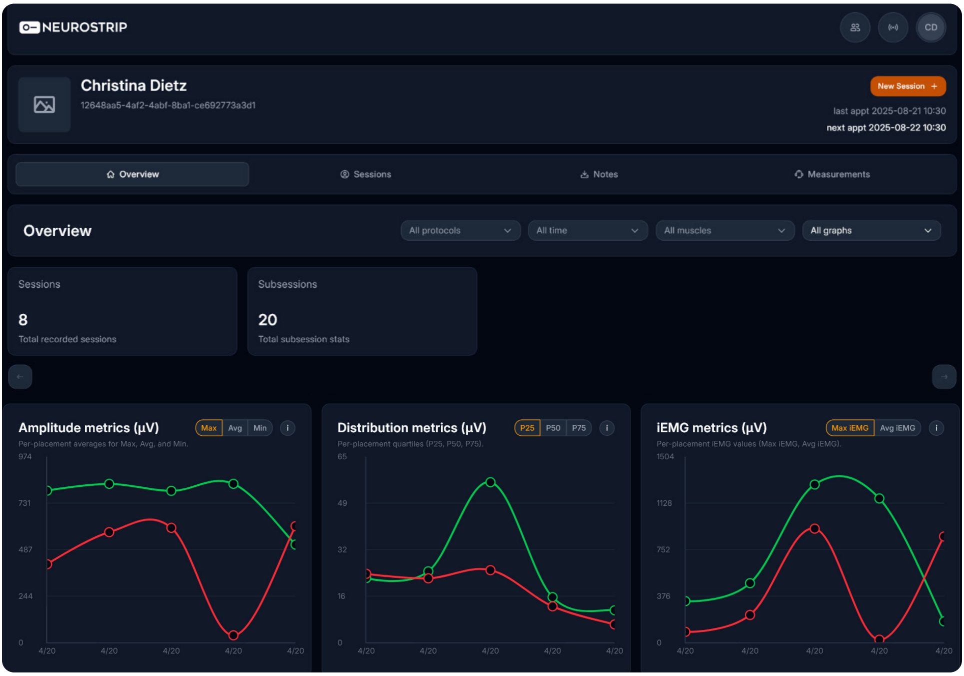
ADD OR SELECT A CLIENT TO BEGIN:
Click the plus sign to add new patient/client, otherwise, click small box to left of name or on the name to select a previously entered patient/client.





NEW PATIENT:
After selecting **+ New Patient** from the intro screen, the new patient pop-up allows users to enter the patient's information and then select **Continue** to proceed.

Tags are clinician-created categories that can be assigned to patients to make searching and organization easier.



OVERVIEW

When a patient is selected from the main screen, this pop-up appears with session options for that individual. The navigation tabs allow users to move between the athlete's main sections: Overview, Sessions, Notes, and Measurements. The Overview shows total Sessions and Subsessions. Use the chart toggles to switch between available metric views.

AMPLITUDE METRICS

Shows per-placement muscle signal averages, including maximum, average, and minimum amplitude values.

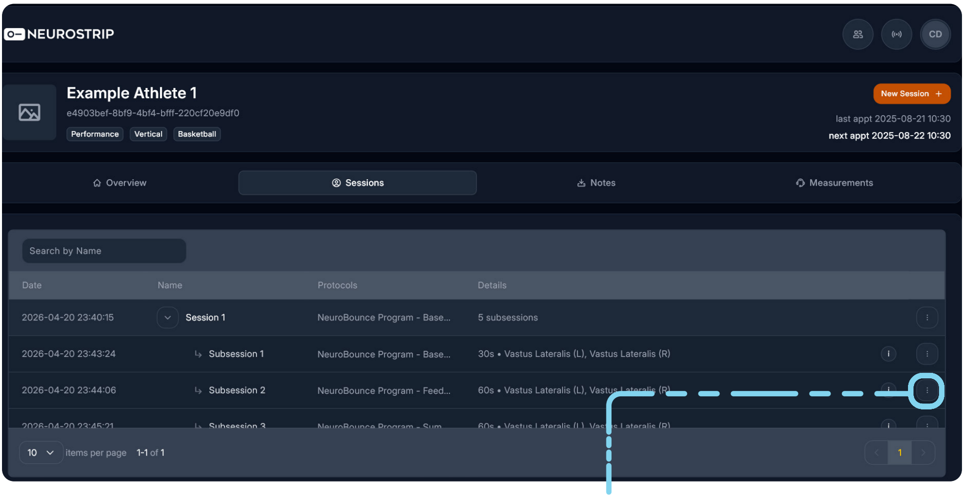
DISTRIBUTION METRICS

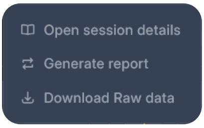
Shows per-placement signal distribution using percentile values, such as P25, P50, and P75.

IEMG METRICS

Shows integrated EMG values, including maximum iEMG and average iEMG.

When ready to perform an exercise, click + New Session to begin.



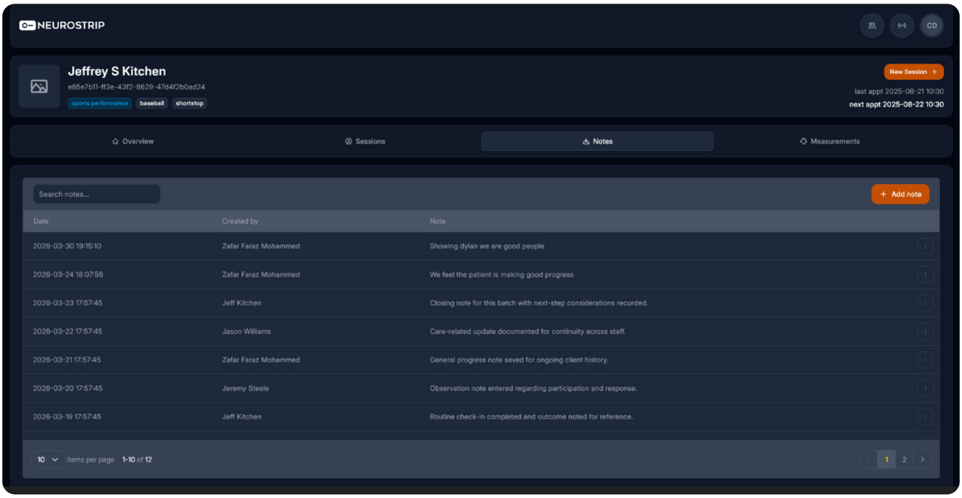


SESSIONS

The Sessions screen is accessed by clicking the Sessions tab. From here, users can review past sessions by double-clicking on a session to open it.

EXPORT OPTIONS

Click dots next to session for session export options.

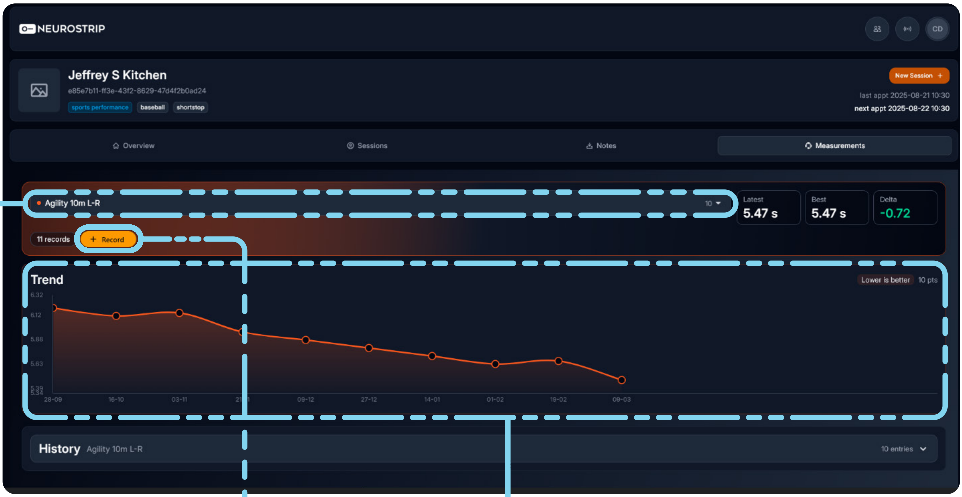


NOTES

The Notes tab displays a searchable list of client notes, including the date, author, and note details, with options to add a new note, view additional note actions, and navigate through saved entries.

MEASUREMENTS TAB

The Measurements tab displays client performance measurements, including the selected measurement type, number of saved records, latest result, best result, performance delta, trend graph, and measurement history, with options to record a new measurement and review progress over time.



MEASUREMENT SELECTOR

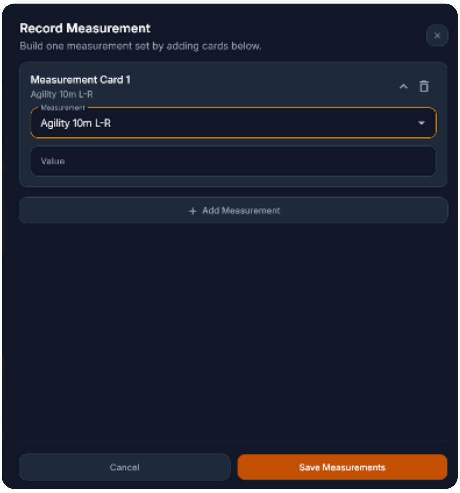
Shows which performance test or measurement is being recorded, such as Agility 10m L-R.

RECORD NEW MEASUREMENT

Click to record new measurement.

TREND GRAPH

Displays performance changes across recorded measurements. In this example, the note "Lower is better" indicates that decreasing values reflect improvement.



RECORD MEASUREMENT

The Record Measurement window allows users to add one or more measurement entries by selecting a measurement type, entering the result value, adding additional measurement cards if needed, and saving the measurements to the client profile.

VALUE FIELD

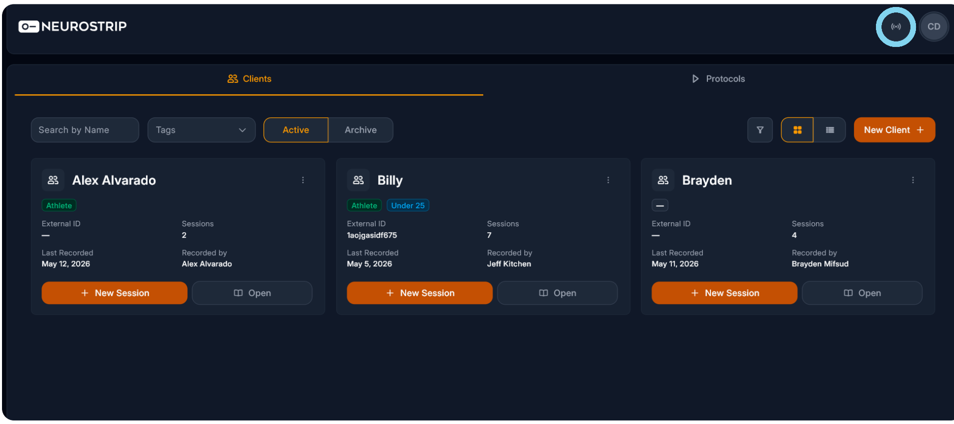
Where the user enters the measurement result before saving.

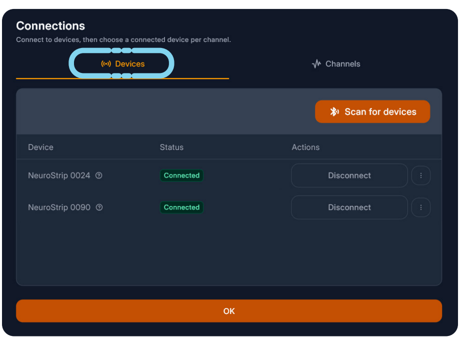
ADD MEASUREMENT BUTTON

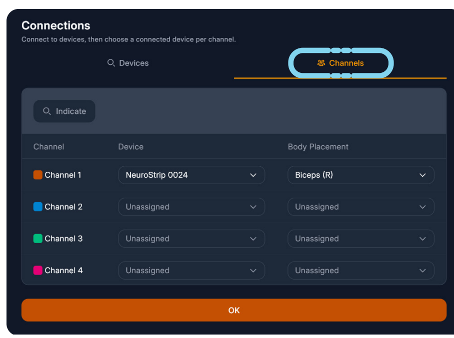
Allows the user to add another measurement card and record multiple measurements at once.

5.0 CONNECTIONS SCREEN

The Connections screen is accessed by clicking the **Connections** tab in the top-right corner of the intro screen. After shaking the device to wake it up, users can select Scan for Devices to locate devices, use the Channels tab to assign devices to a channel and muscle, and click the **?** icon next to the device name to view the current firmware and device serial number.







CONNECTION SCREEN DEVICES

Reached by clicking Connections tab on top right corner of the Intro Screen.

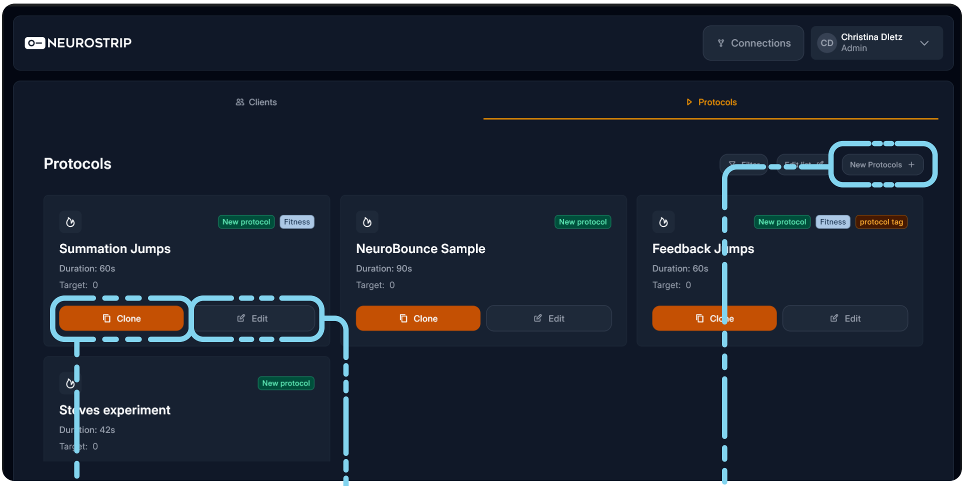
Click **Scan For Devices** after shaking device to wake up. **? icon** next to device name can be clicked to view current firmware and the device serial number.

CONNECTION SCREEN CHANNELS

Able to assign devices to channel and muscle.

6.0 PROTOCOLS SCREEN

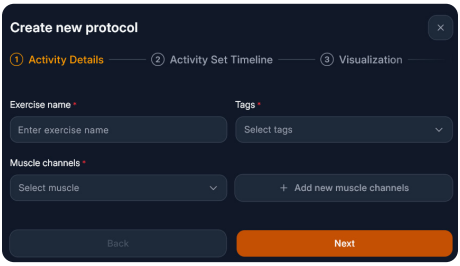
The Protocols screen is accessed by clicking the Protocols tab from the intro screen. From here, users can select Clone to copy an existing protocol or click New Protocol to create a new one.



CLONE
Copy an existing protocol.

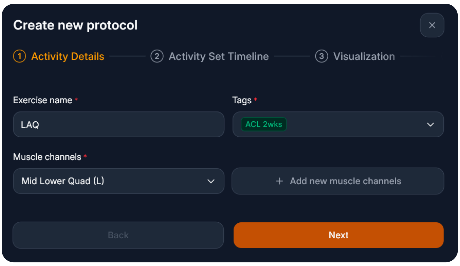
EDIT
Edit an existing protocol.

NEW PROTOCOLS +
Create a new protocol.



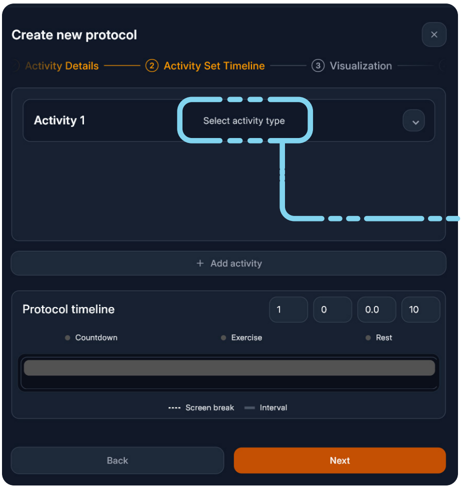
CREATE NEW PROTOCOL

Reached by clicking Connections tab Enter details. Tags can be assigned for easier searching (Examples: ACL rehab, Return to Play, Baseball, etc)



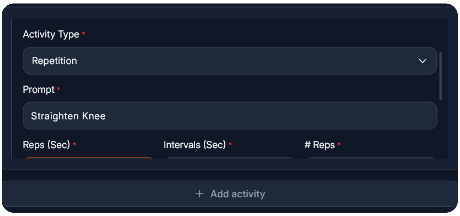
ACTIVITY DETAILS COMPLETE

Completed screen. Select **Next** once filled in.

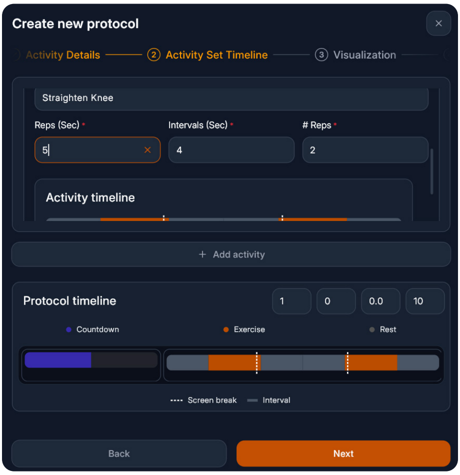


ACTIVITY SET TIMELINE

Select activity type.

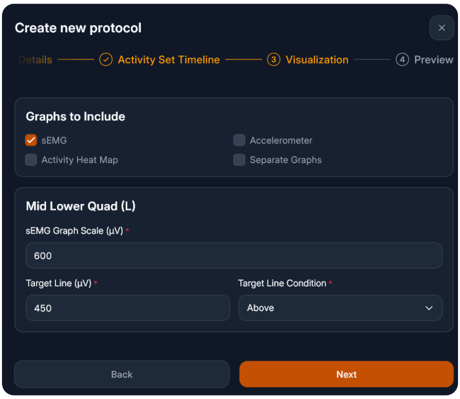


6.1 CREATE NEW PROTOCOL



ACTIVITY SET TIMELINE COMPLETE

Completed screen. Bars below show a preview of what each Activity will look like on screen. Select **Next** once completed.



VISUALIZATION

Graph details can be edited here.

GRAPH SCALE

Max value on Y axis.

TARGET LINE

What patient needs to “hit” for a successful rep.

TARGET ABOVE CONDITION

Rep is successful if rep is above or below target line. Rep will almost always be above.

Click **Next.**

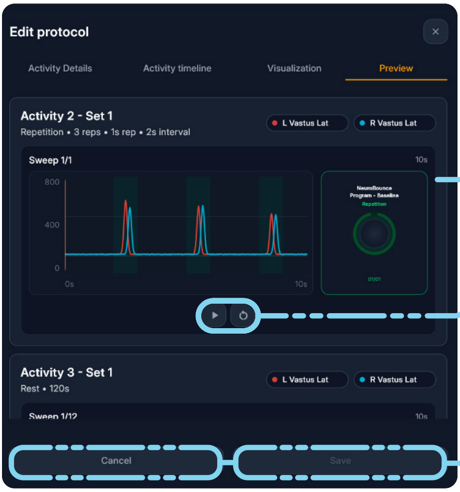
6.2 EDIT PROTOCOL

The Preview tab allows you to review how the protocol will appear and run during a session before saving your changes.

PREVIEW TAB

Each activity is displayed in order, including repetitions, sets, and timing details. For exercise activities, the preview shows the selected muscle channels and the expected activation pattern for each repetition.

The graph displays the timing of each muscle signal during the activity. Each colored line represents a selected muscle channel, such as L Vastus Lat and R Vastus Lat. Peaks on the graph indicate periods of muscle activation, while shaded areas represent the active intervals within the repetition sequence.



SESSION PREVIEW PANEL

The session preview panel on the right shows how the activity will appear to the user during the session, including the program name, activity type, and repetition count.

PLAYBACK CONTROLS

Use the playback controls below the graph to preview or restart the activity. This allows you to confirm that the timing, repetition count, intervals, and visual feedback are correct before saving the protocol.

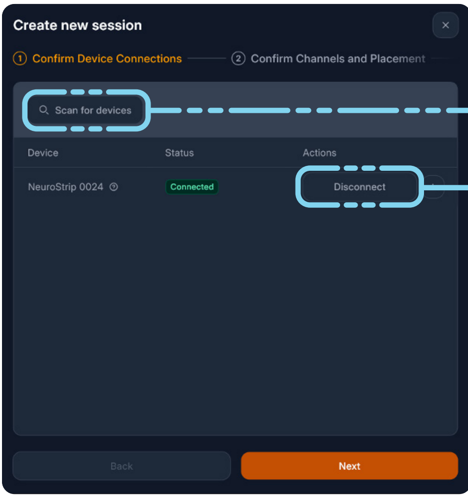
SAVE OR CANCEL

Select **Save** to apply the protocol changes, or select **Cancel** to exit without saving.

7.0 CREATE NEW SESSION

The Create New Session workflow allows users to connect a NeuroStrip device, assign active channels, confirm body placements, and continue to protocol selection.

CONFIRM DEVICE CONNECTIONS



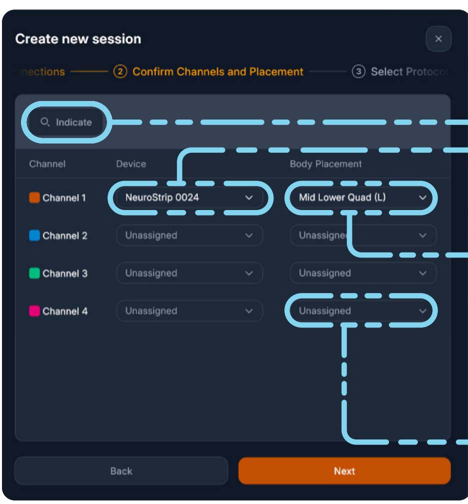
SCAN FOR DEVICES BUTTON
Searches for available NeuroStrip devices that can be connected.

DEVICE LIST
Displays available or connected devices, such as NeuroStrip 0024.

STATUS COLUMN
Shows whether the device is connected. A green Connected status confirms the device is ready.

CONNECT / DISCONNECT ACTION
Allows the user to connect to an available device or disconnect from a currently connected device.

CONFIRM CHANNELS AND PLACEMENT



INDICATE BUTTON
Helps identify or confirm the active device/channel before placement is assigned.

CHANNEL LIST
Displays each available channel, with a unique color indicator for easy identification.

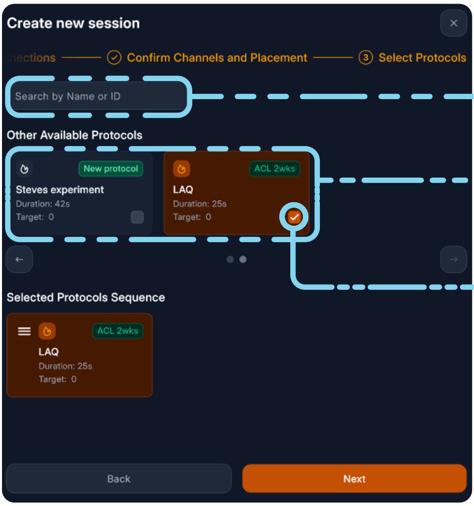
DEVICE DROPDOWN
Assigns each channel to the connected NeuroStrip device.

BODY PLACEMENT DROPDOWN
Selects where the channel is placed on the client's body, such as Mid Lower Quad (L).

UNASSIGNED CHANNELS
Channels that are not being used can remain marked as Unassigned.

SELECT PROTOCOLS

Select protocol by clicking box and then select Next.



SEARCH FIELD

Find a protocol by name or ID.

OTHER AVAILABLE PROTOCOLS

Displays protocols that can be added to the session.

PROTOCOL CARDS

Show key protocol details, including name, duration, target, and protocol label.

SELECTION CHECKBOX

Indicates which protocols are selected for the session.

SELECTED PROTOCOLS SEQUENCE

Shows the order of protocols that will run during the session.

8.0 SESSION GRAPH RECORDING SCREEN

The Session Recording screen allows users to view live EMG activity, monitor channel live readings, track recruitment patterns, and control the recording while a session is in progress. To simplify the graph view, users can turn each EMG visual on or off by selecting the corresponding channel indicator, which is especially helpful when multiple channels are active.

EMG CHART

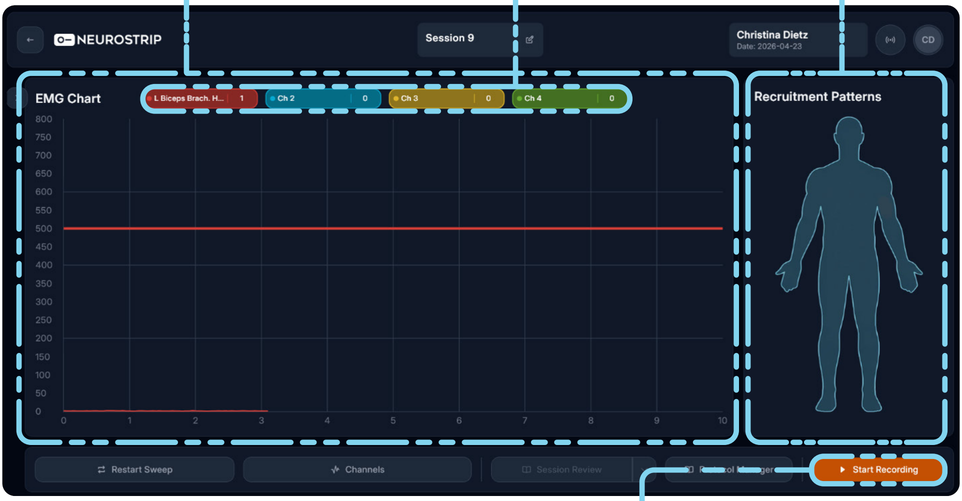
Displays live muscle activity for the active channels during the session.

CHANNEL INDICATORS

Show each assigned channel, its color, selected muscle placement, and current reading.

RECRUITMENT PATTERNS PANEL

Displays a body map visualization for reviewing muscle recruitment patterns during the session.

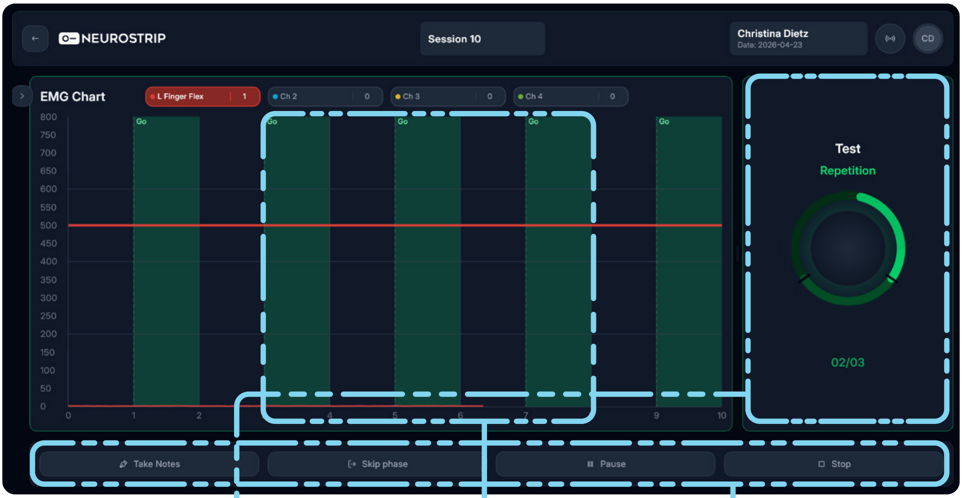


START RECORDING BUTTON

Begins recording data for the selected session protocol.

WIDGET CONTROLS

Allows users to choose which graph elements are visible during the session, including live readings, accelerometer data, raw X/Y/Z data, the EMG chart, and body heatmap; the Audio toggle enables a beep when the target line is reached.



ACTIVE PROTOCOL DISPLAY

Shows the current activity or phase, such as Test and Repetition, during the recording.

REPETITION COUNTER

Tracks progress through the current activity, such as 02/03 repetitions completed.

GO INTERVALS

Green shaded areas on the chart show when the user should perform the activity.

SESSION CONTROLS

Allow users to take notes, skip a phase, pause the session, or stop the recording.

9.0 TROUBLESHOOTING

Below are issues that may arise when using NeuroStrips and the NeuroStrip Application.

CONNECTIVITY ISSUES:

Issue: Devices do not appear when the Connections screen loads.

If devices are not on and in BLE broadcast mode (flashing blue light) prior to moving to the Connections screen, then the receiving BLE connection will not find them. To find them, simply click the **Discover** button at the top of the Connections screen.

Issue: Devices will not connect to the application.

Occasionally hitting the **Connect** button once or twice is required to establish a connection with a NeuroStrip device. Once connected devices will remain connected until the **Disconnect** button is hit.

SEMG GRAPH ISSUES:

Issue: Inaccurate / Erratic readings on the sEMG graph

- Ensure good skin preparation and adherence of the NeuroStrip patch to the body. An overdressing should be applied to ensure adequate pressure on the tail section. The overdressing should be only applied to the tail section or the entirety of the NeuroStrip. Avoid placing an overdressing on the tail and flexible mid-section at the same time. This could cause damage to the mid-section

- Check alignment of the NeuroStrip electrodes to the NeuroStrip patch electrodes. Poor alignment can lead to inaccurate readings. Using the Patch Alignment Tool when placing the NeuroStrip on the patch should eliminate most alignment issues.

- Check sEMG accuracy by placing a connected device into the NeuroStrip charger. When inside the cradle, the NeuroStrip should read near 0μV on the graph. This is an indication that the device is working properly and that there is a likely issue with the physical medium.

- In some scenarios, a hard reset of the NeuroStrip is needed. To do this, use the pin tool found within the kit and insert into the pinhole on top of the head of the NeuroStrip. Push and hold the button down for three seconds to reset the device.

- As a final step for troubleshooting, restart the NeuroStrip Application.

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NEUROSTRIP
USER GUIDE

CONTROL BIONICS